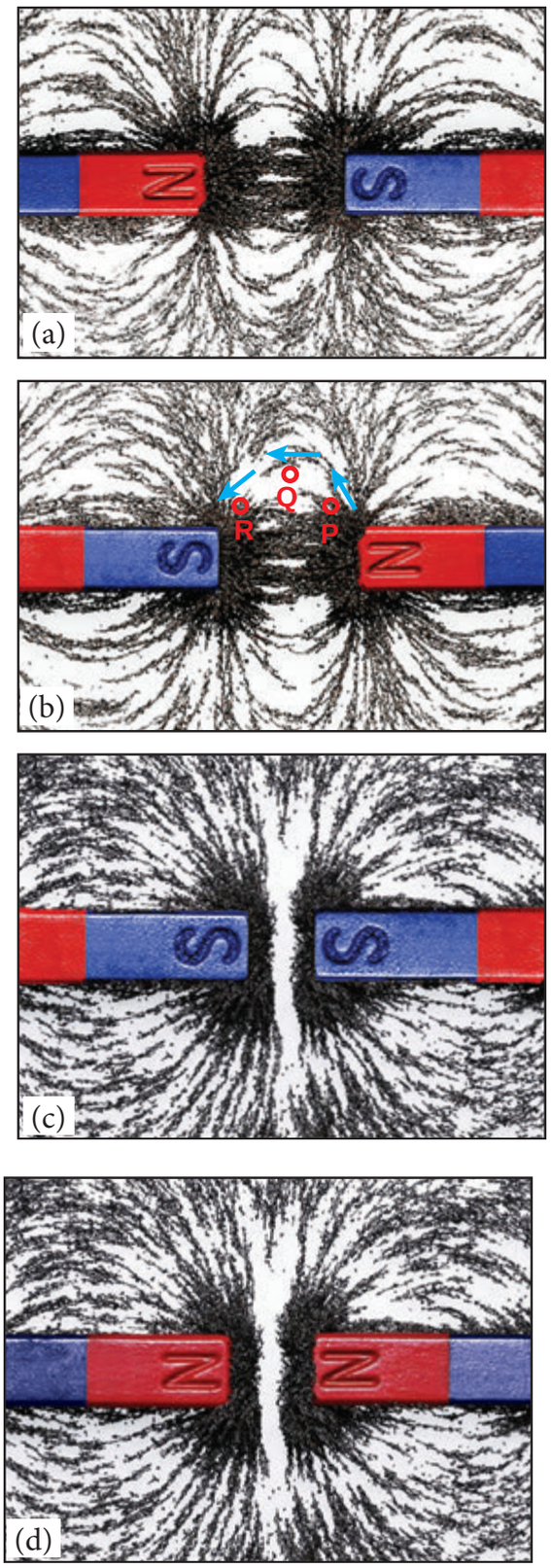



**Figure 3.10** Properties of magnetic field lines– unlike poles attracts each other shown in picture (a) and (b) like poles repel each other-shown in picture (c) and (d)

$$\Phi_B = \vec{B}.\vec{A} = BA\cos\theta = B_\perp A \qquad (3.6)$$

where θ is the angle between $\vec{B}$ *and* $\vec{A}$ as shown in Figure 3.11.

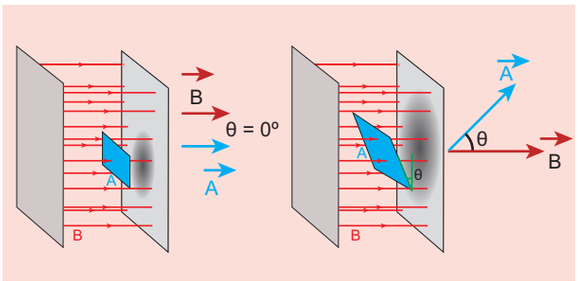



**Figure 3.11** Magnetic flux

### Special cases

(a) When $\vec{B}$ is normal to the surface i.e., θ = 0°, the magnetic flux is $\Phi_B$ = BA (maximum).

(b) When $\vec{B}$ is parallel to the surface i.e., θ = 90°, the magnetic flux is $\Phi_B = 0$.

Suppose the magnetic field is not uniform over the surface, the equation (3.6) can be written as

$$\Phi_B = \int \vec{B}.d\vec{A}$$

Magnetic flux is a scalar quantity. The SI unit for magnetic flux is weber, which is denoted by symbol Wb. Dimensional formula for magnetic flux is $\left[ML^2T^{-2}A^{-1}\right]$. The CGS unit of magnetic flux is Maxwell.

1 weber = $10^8$ maxwell

**The *magnetic flux density* can also be defined as the number of magnetic field lines crossing unit area kept normal to the direction of line of force.** Its unit is Wb $m^{-2}$ or tesla.



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